

Atharvaa Rane

Little Rock, AR | +1 (315) 886 4890 | atharvaarane89@gmail.com | [LinkedIn](#)

SUMMARY

Software Engineer with around 4 years of experience building AI-enabled and data-driven Python applications, with a strong focus on scalability, performance, and production-ready machine learning systems. Experienced in developing backend services and ML pipelines using Python, Django, and Flask, and integrating machine learning and NLP models into enterprise applications. Skilled in data preprocessing, feature engineering, model validation, and ML inference services supporting real-time analytics and decision automation. Proficient in designing and managing relational and NoSQL databases including MySQL, PostgreSQL, SQL Server, MongoDB, and Firebase. Hands-on experience in building RESTful APIs to expose AI/ML capabilities and deploying containerized applications on AWS, Azure, and Databricks.

SKILLS

Methodology: SDLC, Agile, Waterfall

Programming Languages: Python, SQL, PL/SQL, Java, C#, ASP.NET, C, C++

IDES: Visual Studio Code, Jupyter Notebook, PyCharm

Machine Learning & AI: Supervised & Unsupervised Learning, Regression, Classification, Clustering, Time-Series Forecasting, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Generative AI

AI / LLM Models: GPT-OSS, GPT OSS, Open-Source GPT Models, Transformer-Based Models

ML Frameworks & Libraries: Scikit-learn, TensorFlow, NumPy,

Pandas, Web2py, CherryPy, Requests, XGBoost, LightGBM

Natural Language Processing (NLP): Text Classification, Sentiment Analysis, Topic Modeling, Named Entity Recognition

Frameworks: Django, Flask, Node.js, Express.js, React.js, Angular, .Net Framework

Cloud Technologies: AWS, Azure, DataBricks, Docker

Web Technologies: HTML, CSS, JavaScript, AJAX, XML, JSON, jQuery, Bootstrap, Restful API, Redux

Database: MySQL, SQL Server, MongoDB, Firebase, PostgreSQL

Version Control/Other Tools: GitHub, Git, GitLab, Jira

Operating Systems: Windows, Linux, Mac

EXPERIENCE

COSMOS Research Center, University of Arkansas at Little Rock | Aug 2025 - Present

Software Engineer (AI)

- Designed, deployed, and owned production-grade AI applications using Python, integrating machine learning, NLP, and Generative AI (LLMs) to deliver scalable analytics and automation solutions.
- Productionized end-to-end ML pipelines (regression, classification, clustering, time-series forecasting) using Scikit-learn, XGBoost, and LightGBM, improving model accuracy by 20% and supporting repeatable experimentation.
- Implemented LLM-powered services using open-source GPT models to enable text classification, sentiment analysis, and automated insight generation at scale.
- Built and maintained RESTful AI inference services to support real-time predictions and seamless integration with downstream research and data platforms.
- Containerized and operationalized ML and LLM workflows using Docker, improving environment reproducibility and reducing deployment cycle time by 30%.

Honeywell, USA | Nov 2024 - Jul 2025

Software Engineer

- Developed and deployed production AI/ML services supporting underwriting and claims analytics, enabling automated risk scoring and reducing decision turnaround time by 28%.
- Owned the design and deployment of supervised ML models for customer segmentation and policy lapse prediction, improving accuracy by 18% across pilot programs.
- Built robust data preprocessing, feature engineering, and model validation pipelines, increasing data reliability and consistency for training and real-time inference.
- Containerized and deployed ML inference services on AWS, improving deployment stability and reducing release-related incidents by 25%.
- Partnered closely with actuarial, data science, and business stakeholders to translate insurance requirements into scalable, production-ready AI solutions.

Mindtree, India | Jan 2021 - Jul 2023

Software Engineer

- Delivered and supported production AI-enabled Python applications for policy servicing and claims analytics, reducing manual review effort by 30% through automated ML-driven decision workflows.
- Built scalable backend services using Python, Django, and Flask to operationalize ML model outputs within policy administration and reporting systems.
- Designed and exposed high-performance RESTful APIs for predictive insights and ML inference, reducing response latency by 35% and improving downstream consumption.
- Developed data preparation and feature engineering pipelines to standardize inputs for training and production inference, improving model readiness and consistency.
- Led containerization and AWS deployment of ML-enabled applications, improving deployment reliability and ensuring consistent environments across development, testing, and production.

EDUCATION

Master of Science in Computer Science | Syracuse University, NY

May 2025